

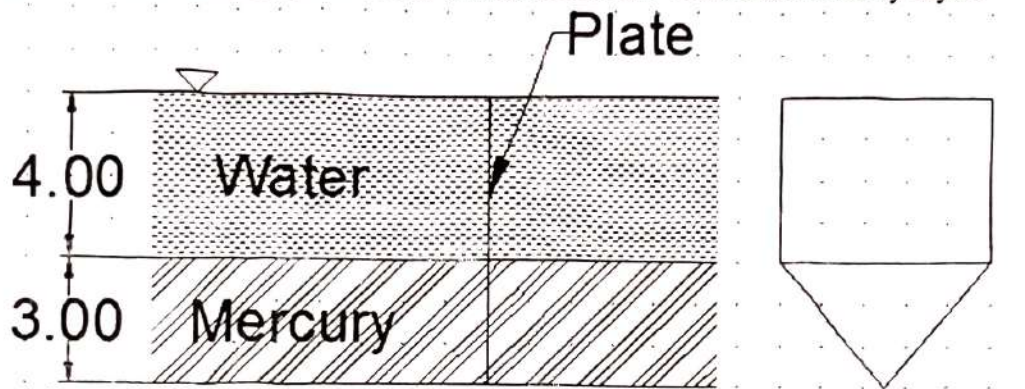
Duration: 45 mins

Max Marks: 20

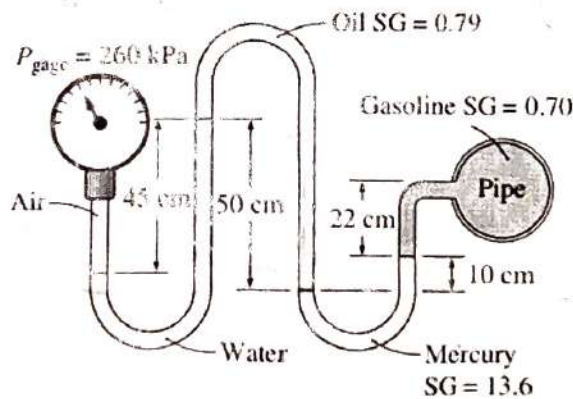
Note: Please write the answers in SI units and up to three decimal points with units. Use $g=9.81 \text{ m/s}^2$.

Q1. A thin plate consists of one rectangular portion and one triangular portion attached below it with the apex of the triangle is the bottom most part of the plate. The size of rectangle is 5 m width and 4 m depth, and triangle base is 5 m and height is 3 m. The plate is submerged **vertically** under liquid layers consisting of 4 m water above mercury depth of 3 m. Neglect the weight of the plate and any reactions and find hydrostatic force and center of pressure from top of the water surface. Specific gravity of Mercury is 13.6.

b) What is the total force on plate if it is **horizontal** at the bottom of mercury layer?



Q2. A gasoline line is connected to a pressure gage through a double-U manometer, as shown in Figure. If the reading of the pressure gage is 260 kPa, determine the gage pressure of the gasoline line. [10]



Q3. A semicircular 40 m diameter tunnel is to be built under water as shown in Figure. Determine the total hydrostatic force acting on the roof of the tunnel if the liquid above it is water. [5]

